

A QUESTIONS IN THE QUIZ DURING 2023-2025

A.1 2023 – Part V: Comprehensive Problem (20 points)

Table A.1: Statistics of 2023 Teaching Database Relations.

Relation	# Tuples	# Physical Pages	Primary Key
Teacher(Tno, Tname, Tsex, Tage, Tdept)	2000	5	Tno
Course(Cno, Cname, Credit, Cppo)	1200	3	Cno
TC(Tno, Cno, Year)	6000	14	(Tno, Cno)

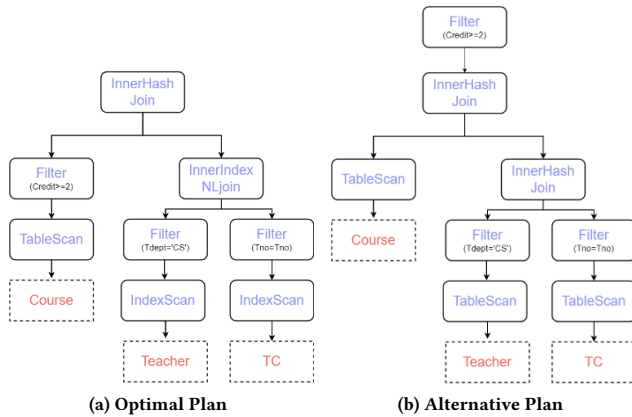


Figure A.1: Optimal Plan and Alternative Plan for the 2023 query.

Consider the teaching database in Table A.1.

Assume:

- TC.Tno is a foreign key referencing Teacher.Tno.
- TC.Cno is a foreign key referencing Course.Cno.
- Teachers are uniformly distributed across Tdept, and TC are uniformly distributed across Year.
- Teacher names are unique.
- Each table has an in-memory B⁺ tree index built on its primary key.
- Teacher.Tdept also has a B⁺ tree index.
- All intermediate results can be kept in memory during query execution.

By the end of 2022, the university must report every course with credit ≥ 2 that the “CS” department teachers offered that year. Determine the qualifying course names and address the following:

- (1) Figures A.1a and A.1b show the optimal plan and an alternative plan for the SQL query. Explain the advantages of the optimal plan over the alternative plan. (10 points)
- (2) Other than the two plans in part (1), describe at least one additional executable plan for requirement (1) and explain why the optimizer might reject it. (10 points)

A.2 2024 – Part V: Query Understanding (20 points)

Table A.2: Statistics of 2024 Advising Database Relations.

Relation	# Tuples	# Physical Pages	Primary Key
Student(id, name, sex, age, dept)	20000	50	id
Instructor(id, name, age, dept)	1000	3	Cno
Advisor(s_id, i_id, Course)	1000000	600	(s_id, i_id)

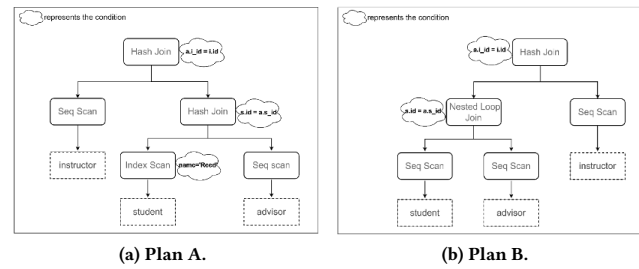


Figure A.2: Optimal Plan and Alternative Plan for the 2024 query.

Consider the database in Table A.2. The three relations correspond to student information, instructor information, and advising relationships, respectively.

Assume:

- Advisor.s_id is a foreign key referencing Student.id.
- Advisor.i_id is a foreign key referencing Instructor.id.
- Students are uniformly distributed across dept.
- Student names are unique.
- Each table has an in-memory B⁺ tree index built on its primary key.
- “Seq Scan” denotes a table scan operator.

Consider the query below.

```
SELECT s.name, i.name
FROM student AS s, advisor AS a, instructor AS i
WHERE s.name = 'Reed'
AND s.id = a.s_id
AND a.i_id = i.id;
```

Two alternative plans for this query are shown in Figures A.2a and A.2b, and plan A is the optimizer’s choice.

- (1) List possible reasons why plan A is superior to plan B. (10 points)
- (2) Sketch a third potential plan (plan C) for this query and explain why the optimizer is unlikely to choose it. (10 points)

A.3 2025 – Part V: Query Understanding (20 points)

Table A.3: Statistics of 2025 Teaching Database Relations.

Relation	# Tuples	# Physical Pages	Primary Key
Course(credits, course_id, title, dept_name)	100	2	Course_id
Instructor(id, name, age, dept)	1000	10	id
Teaches(year, id, course_id, sec_id, semester)	60000	600	(s_id, i_id)

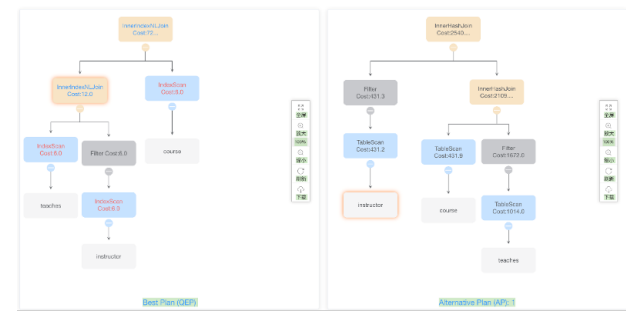


Figure A.3: Optimal Plan and Alternative Plan for the 2025 query.

Consider the database in Table A.3, where the three relations store course information, instructor information, and teaching relationships.

- Assume:
- Teaches.id is a foreign key referencing Instructor.id.
 - Teaches.course_id is a foreign key referencing Course.course_id.
 - Instructors are uniformly distributed across dept.
 - Instructor names are unique.
 - Each table has an in-memory B⁺ tree index built on its primary key.
 - “Seq Scan” denotes a table scan operator.

Consider the SQL query:

```
-- List courses Professor Lembr teaches in the Fall semester.
SELECT title, semester
FROM course, teaches, instructor
WHERE course.course_id = teaches.course_id
      AND teaches.id = instructor.id
      AND instructor.name = 'Lembr'
      AND teaches.semester = 'Fall';
```

- (1) Figure A.3 depicts the optimal and an alternative plan. Analyze the advantages the optimal plan has over the alternative plan. (10 points)
- (2) Describe at least one additional plan that satisfies the query and explain why it might not be chosen for execution. (10 points)